

# SENTHIL NAGAPPAN

## AI Experience Design & Human Systems Integration

Accessibility Leadership · Design Systems · Production Front-End · Safe, Human-Centered AI in Regulated Environments  
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## PROFILE

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AI experience designer and human-systems-integration leader with 18+ years pairing accessibility leadership, design systems, and production front-end engineering to deliver safe, human-centered AI across healthcare, federal, and enterprise environments. I design the experiences and the strategic logic that let human-centered empathy govern intelligent, non-deterministic systems — building trust, transparency, and control into agentic and data-rich products, then prototyping and shipping them in real code.

What sets my work apart is the rare combination of ends I hold together: the research and human-factors discipline to know what to build, the accessibility and design-system rigor to make it inclusive and consistent at scale, and the front-end engineering fluency to build and ship it myself. I move ideas from concept to production without losing craft or clarity — and I do it in regulated, high-stakes domains where getting the human experience right genuinely matters. I lead and mentor design teams, partner deeply with product and engineering, and translate complex, ambiguous problems into clear, buildable, trustworthy solutions.

## WHAT I DO

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**AI Experience Design** Agentic & human-AI collaboration, human-in-the-loop control, trust / transparency / explainability, designing for uncertainty and non-deterministic outputs, human-centered governance of AI, multimodal experiences

**Human Systems Integration** Human factors & usability engineering, cognitive-load & decision-support design, summative & formative research, research synthesis, safety-critical and regulated-product experience (medical devices, clinical software, FDA HFE / IEC 62366)

**Accessibility Leadership** WCAG 2.2 AA, Section 508, ADA, inclusive design, assistive-technology fluency (JAWS, NVDA, VoiceOver, TalkBack, Dragon), accessibility embedded into design systems by default — IAAP CPACC + WAS (CPWA)

**Design Systems** Component-library architecture & governance, design tokens, interaction & accessibility patterns, contribution & versioning models, consistency at scale across multiple independent teams

**Production Front-End** React / TypeScript, Redux, HTML / CSS / JavaScript, design-to-code & vibe-coding (Lovable, Cursor, v0, Figma Make), building and shipping working prototypes, PWAs, and production UI

**Research & Strategy** Usability testing, interviews, contextual inquiry, heuristic evaluation, data synthesis, journey mapping, problem framing, roadmap influence, executive communication & design leadership

**Craft & Tools** Figma (expert), FigJam, Sketch, Adobe Creative Suite, brand identity, prototyping & motion, data visualization & dashboards; AI / LLM prompt engineering and daily GenAI practice

## SELECTED EXPERIENCE

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### AI Experience Designer & HSI Lead — GE HealthCare (Brain Pulse Inc) Mar 2025 – Present

*Agentic clinical workflows · human-AI trust & explainability · design-to-code · design systems*

- Design end-to-end experiences for clinical products — from agentic-workflow logic and human-in-the-loop control to high-fidelity flows — so clinicians can understand, calibrate, and trust intelligent, probabilistic systems in high-stakes decisions.
- Design for trust, transparency, and explainability — applying human-in-the-loop patterns and user-control affordances that keep the human in command of non-deterministic AI outputs.
- Bridge AI and UI through design-to-code — vibe-coding working prototypes (Lovable, Cursor, v0) and production front-end (React/TypeScript) so design and engineering move as one integrated team.
- Build and govern reusable design-system components adopted across 10+ teams, embedding accessibility and human-factors rigor so safe, human-centered design is structural rather than an afterthought.

### Sr. UX & Accessibility Lead — Social Security Administration (Leidos) Oct 2022 – Feb 2025

*Enterprise design system · accessibility platform · research-driven governance · 50M+ users*

- Architected and governed an enterprise design system (component library, tokens, interaction patterns) used simultaneously by multiple independent development teams across 15+ modules serving 50M+ users.
- Drove research and evidence-based design — traced 65% of interaction issues to just 8 reused components, then standardized them at the source, driving governance decisions and saving \$1.5M+ annually.
- Made WCAG 2.2 AA / Section 508 accessibility a built-in property of the system, delivered across 40+ monthly releases with seamless design-to-development hand-off.
- Led and mentored designers and partnered with senior stakeholders — translating complex requirements into clear, buildable system design and defending design decisions with evidence.

### Sr. UX & Inclusive Design Lead — Best Buy Health

Sep 2021 – Sep 2022

*Connected medical devices · summative validation · native mobile & web · 1M+ users*

- Designed connected-care and medical-alert experiences for 1M+ users, including older adults (65–90+), across native mobile (iOS/Android) and web, aligned to a shared design system.
- Led summative usability validation with critical tasks derived from hazard analysis and FMEA, contributing to 510(k) human-factors evidence and driving safety-critical design changes.
- Built and governed the product design system (BRIX) — reusable components and patterns that improved cross-product consistency and reduced remediation cycles 30–40%.

### **Sr. UX & Program Lead — HHS / SAMHSA (Eagle Technologies) Mar 2014 – May 2021**

*Behavioral-health data platforms · data visualization · design standards · 2M+ users*

- Designed enterprise web and data-visualization tools across a large behavioral-health portfolio serving 2M+ users, translating complex data into meaningful, actionable experiences.
- Established reusable components and design standards across programs, reducing rework 30–40% (\$2M+ saved) and mentoring designers over a multi-year, multi-team engagement.

**Earlier:** Lead UX Designer — AAMC · Sr. UX & Accessibility Consultant — CareFirst BCBS · UX Designer — JPMorgan Chase · UX Consultant — Morgan Stanley (financial services, healthcare & enterprise application design).

### **SELECTED AI & PROTOTYPING WORK**

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- Full-stack AI-powered visual-assistance app for blind and low-vision users (inspired by Be My Eyes / OrCam) — designed and built end-to-end as a Vite + React PWA with serverless AI-proxy functions, featuring real-time scene description, OCR text reading, and product identification. A working demonstration of AI experience design and design-to-code fluency in one project.
- Enterprise AI-experience pattern library — explainability, trust calibration, agentic hand-off, and human-in-the-loop control patterns applied across clinical and connected-product work.
- Interactive design-system and data-visualization prototypes — production-caliber front-end work showing how design systems scale and how complex data becomes clear, actionable insight. Full case studies at [www.senthilnagappan.com](http://www.senthilnagappan.com).

### **EDUCATION & CERTIFICATIONS**

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B.E., Electronics & Communication Engineering (Human-Computer Interaction / Human Factors focus) — Madurai Kamaraj University, 1999.

IAAP CPACC + WAS (CPWA — Accessibility) · CUA (Certified Usability Analyst) · CSM (Scrum Alliance — Agile) · Salesforce Certified UX Designer (in progress).

### **LEADERSHIP & COMMUNITY**

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Vice President, Texas Astronomical Society — lead organizational strategy and volunteer coordination for a 500+ member nonprofit, with a focus on accessible astronomy outreach.

Cubmaster & Assistant Scoutmaster, Scouting America (Circle Ten Council) — Wood Badge trained; recipient of the Spark Plug Award (2025), Iron Horse District.

Lead Coach, FIRST LEGO League — coach youth robotics teams through the engineering design process.